Algebraic Groups

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1 Algebraic groups

Definition 1.1. Let be G an algebraic variety, $\phi : G \longrightarrow G$ and $\psi : G \times G \longrightarrow G$ regular maps such that for any $g, h, t \in G$:

- $\psi(\psi(g, h), t)) = \psi(g, \psi(h, t))$ (*associativity*);
- $\exists e \in G \mid \psi(e, g) = g = \psi(g, e)$ (*existence of a neutral element*);
- $\psi(g, \phi(g)) = e = \psi(\phi(g), g)$ (*existence of inverses*).

Then G is a group and we will call it algebraic group. We will write $g^{-1} = \phi(g)$ and $gh = \psi(g, h)$.

Example 1.2. Let's see some examples:

- The affine line $\mathbb{A}^1$ with the group aw defined by the maps $\psi(a, b) = a + b$ and $\phi(a) = a$ is called the additive group and is denoted by $\mathbb{G}_a$;
- The variety $\mathbb{A}^1 \setminus \{0\}$ with the group law defined by the maps $\psi(a, b) = ab$ and $\phi(a) = \frac{1}{a}$ is called multiplicative group and is denoted $\mathbb{G}_m$.
- Let us consider a cubic plane curve X, by theorem on shafarevich [1] we have that the maps $\phi : X \to X$ $\phi(a) = \ominus a$ and $\psi : X \times X \to X$ $\psi(a, b) = a \oplus b$ are regular. Since these maps follow the rules above we have that a plane cubic curve with this operation and these inverse is an algebraic group.

Theorem 1.3. An algebraic group is a nonsingular algebraic variety.

Proof. Let be G an algebraic group, it follows from the regularity of ψ that for any $h \in G$ $t_h := \psi(h, \cdot) : G \longrightarrow G$ is regular, moreover we also have that $(t_{h^{-1}} \circ t_h)(g) = t_{h^{-1}}(hg) = h^{-1}hg = g$. Then we have that t_h is bijective, regular with a regular inverse, so is an automorphism for any $h \in G$.
For any $a, b \in G$ exists $c := ba^{-1}$ such that $t_c(a) = ca = ba^{-1}a = b$; we also have that the property that a point is singular is invariant under isomorphism[2]. Absurdly if exists one

[1] 3.12 pag 176
[2] Lemma 5.3.8 Caporaso

singular point, then all the point of G will be singular. But this would contradicts the fact that the set of singular points in a variety is a proper subset of the variety[3]. Then G has no singular point. □

We can generalize the above results: suppose to have a variety X with his group G of automorphism, with hte property that for any two point $x_1, x_2 \in X$ exists $g \in G$ such that $g(x_1) = x_2$. In this case we say that X is *homogeneous*. The upper argument shows that a homogeneous variety is nonsingular.

2 Abelian Varieties

Definition 2.1. An abelian variety is an algebraic group G such that is also a projective and irreducible variety.

Definition 2.2. A family of maps between varieties X and Z with base Y an algebraic variety is a map $f : X \times Y \to Z$. Obviously for any $y \in Y$ we have a map $f_y : X \to Z$ defined by $f_y(x) := f(x, y)$.

Now we are going to state 3 results in order to prove a lemma.

Lemma 2.3. Given X a projective variety and Y a quasiprojective variety, then the projection $p : X \times Y \to Y$ is a closed map.

Lemma 2.4. Let $f : X \to Y$ be a regular map between irreducible varieties. Suppose that f is dominant and $dimX = n$ and $dimY = m$. Then $m \leq n$ and $dimF \geq n - m$ fora any $y \in Y$ and for any component F of the fibre $f^{-1}(y)$.

Lemma 2.5. Let be X an irreducible variety and $Y \subsetneq X$ closed, then $dimY < dimX$. [4]

From the projectivity condition, follow a lot of interesting properties.

Lemma 2.6. Suppose that X and Y are two irriducible varieties with X projective, and let $f : X \to Y \to Z$ be a family of regular maps from X to a variety Z with base Y. Suppose that exists at least one point $y_0 \in Y$ such that $f(X \times \{y_0\})$.

Proof. Consider the graph of f $\Gamma := \{(x, y, z) \in X \times Y \times Z | f(x, y) = z\}$. We have that $\Gamma \subset X \times Y \times Z$ and $\Gamma \cong X \times Y$[5]. If we consider the projection $p : X \times Y \times Z \to Y \times Z$ such that $(x, y, z) \mapsto (y, z)$, then we have $p(\Gamma) = p(\{(x, y, z) \in X \times Y \times Z | f(x, y) = z\}) = \{(y, z) \in Y \times Z | \exists x \in X$ such that $f(x, y) = z\}$. Since X is projective and Γ is closed, by lemma 2.3 we have that $p(\Gamma)$ is closed in $Y \times Z$.
Now we can consider the following map $q : p(\Gamma) \to Y$ such that $(y, z) \mapsto y$, the restriction of another projection. Now let $y \in Y$, we now that forall $x \in X$ exists $z \in Z$ such that $f(x, y) = z$. Since $X \neq \emptyset$ we have that the fiber of q over y are $q^{-1}(y) = \{(y, z) \in Y \times Z | \exists x \in X$ such that $f(x, y) = z\} = \{y\} \times f(X \times \{y\})$ then we have that q is surjective.

[3]Pag 102 proposition 5.4.4 of Caporaso.
[4]Pag 58 proposition 3.14.5 of Caporaso.
[5]The isomorphism is $h(x, y) \mapsto (x, y, f(x, y))$ and the projection is its inverse.

We also know that exixsts $z_0 \in Z$ such that $f(X \times \{y_0\}) = z_0$. Moreover we have that $p(\Gamma)$ is irriducible since is immage of Γ, that is irriducible since is isomorphic to $X \times Y$, thorugh the map p that is a projection, and then is continuos. Since q is the restriction of the projection we have that is regoular, then we can aplly the lemma 2.4 and we have that $0 = dim(y_0, z_0) = dim(q^{-1}(y_0)) \geq dim(p(\Gamma)) - dimY \geq 0$ then we have that $dim(p(\Gamma)) = dimY$.
Now let $x_0 \in X$, we have that $W_{x_0} := \{(y, f(x_0, y)) | y \in Y\} \subset p(\Gamma)$ is a variety, closed and is isomorphic to Y. Since Y is irreducible, W_{x_0} is closed and they have the same dimension they must be equal by lemma 2.5, and therefore $\forall x_1, x_2 \in X$, $W_{x_1} = W_{x_2}$ and then $y \in Y$ $f(x_1, y) = f(x_2, y)$. The lemma is proved. □

Remark 2.7. Without the assumption that X is projective, the lemma is false: let be $f_y : \mathbb{A}^1 \to \mathbb{A}^1$ such that $f(a, b) := ab$ a family of regular maps, $\mathbb{A}^1$ is not projective, then we can't use the lemma 2.3 and in fact we have that $p(\Gamma)$ is not closed and then we can't apply the lemma 2.4. We have that $p(\Gamma) = \{(y, z) \in \mathbb{A}^2 | \exists x \in \mathbb{A}^1$ such that $xy = z\} = \mathbb{A}^2 \setminus \{(y, z) \in Y \times Z | y = 0$ and $z \neq 0\}$, and lemma 2.4 is false because $0 = dim(0, 0) = dim(q^{-1}(0)) \geq dim(p(\Gamma)) - dim(\mathbb{A}^1) = 2 - 1 = 1$.

Theorem 2.8. An abelian variety is an abelian group.

Proof. We can consider the family of maps from G to G with base G given by $f(g, h) = g^{-1}hg$. When $h = e$ we have that $f(g, e) = e$ so by the lemma 2.6 we have that $f(G \times \{h\})$ is a single point in G hence for any $g_1, g_2 \in G$ $f(g_1, h) = f(g_2, h)$. Hence for any $g, h \in G$ we have $g^{-1}hg = f(g, h) = f(e, h) = ehe = h$, so $hg = gh$, and this means that is abelian. □

Theorem 2.9. Let G an abelian variety, H an algebraic group and $\alpha : G \longrightarrow H$ a regular map. Then exists $\beta : G \longrightarrow H$ group homomorphism such that $\forall g \in G$ $\alpha(g) = \alpha(e)\beta(g)$ where e is the identity element of the group G.

Proof. We define $\beta : G \longrightarrow H$ as $\beta(g) := \alpha(e)^{-1}\alpha(g)$, and we only have to prove that β is a group homomorfism. We can consider the following family of maps $G \longrightarrow H$ with base G:

$$f : G \times G \longrightarrow H \text{ such that } f(h, g) = \beta(h)\beta(g)\beta(hg)^{-1}$$

Since $\beta(e) = \alpha(e)^{-1}\alpha(e) = e' \in H$ is the identity element of H we have that $f(g, e) = \beta(g)\beta(e)\beta(ge)^{-1} = \beta(g)\beta(g)^{-1} = e'$, so $f(G \times \{e\}) = \{e'\} \in H$ is a point. By lemma 2.6 $f(G, g)$ is a single point for every $g \in G$, so for any $g_1, g_2 \in G$ $f(g_1, g) = f(g_2, g)$. Hence we have $\beta(g_1)\beta(g)\beta(g_1g)^{-1} = f(g_1, g) = f(e, g) = \beta(e)\beta(g)\beta(eg)^{-1} = \beta(g)\beta(g)^{-1} = e'$ and so $\beta(g_1)\beta(g) = \beta(g_1g)$ for any $g, g_1 \in G$, so β is a group homomorphism. □

Corollary 2.10. Two abelian varieties isomorphic as algebraic varieties, are also isomorphic as groups.

Proof. Let be G, H abelian varieties with $\alpha : G \longrightarrow H$ isomorphism. By theorem 2.9 exists $a : G \longrightarrow H$ group homomorphism such that $a(g) = \alpha(e)^{-1}\alpha(g)$. We can also apply the theorem 2.9 to α^{-1}, hence exists a group homomorphism $b : H \longrightarrow G$ such that $b(h) =$

$(\alpha^{-1}(e'))^{-1}(\alpha^{-1}(h))$. Then $a \circ b(h) = a((\alpha^{-1}(e'))^{-1}(\alpha^{-1}(h))) = a((\alpha^{-1}(e')^{-1})a(\alpha^{-1}(h))) =$[6] $(\alpha(e)^{-1}\alpha(\alpha^{-1}(e')))^{-1}\alpha(e')\alpha(\alpha^{-1}(h)) = (\alpha(e)^{-1}e')^{-1}\alpha(e)^{-1}h = (\alpha(e)^{-1})^{-1}\alpha(e)^{-1}h = \alpha(e)\alpha(e)^{-1}h = h$. Analogously we can prove that $b \circ a(g) = g$, then we have that a is an isomorphism of group. $\square$

[6]For all homomorfism's group f we have that $f(g^{-1}) = (f(g))^{-1}$.